

Socially Distributed Cognition : Chipotle

Michael Chow
Info 2131 , 2/9/25 - 2/12/25

Overview of Project page 1

Why did I choose this site?

I choose Chipotle because it is very close to campus and I have been there many times before. Chipotle is also a great place for this type of project because we can see almost everyone that works there and see the process that it takes to do this project.

What did I expect to see here in terms of Information Science?

Walking in, I had a strong sense of what I was going to see and what I needed to do in order to be part of the ecosystem. Like most restaurants, I expected to be the customer and go in and order food to some employees. I tried to change my mindset to be a new customer walking in for the first time and grasp the full feel of the project.

What do they “produce”?

Chipotle produce a wide range of american-mexican foods such as burritos, burrito bowls, quesadillas, tacos and a wide variety of drinks. Chipotle lets you decide what you want to put in these foods with different varieties of meats, vegetables and condiments.

Overview of Project page 2

What task did I observe

When eating at Chipotle, I observed mine and other customer's food being created by the staff. Not only do we see employees working at the "assembly line" and cashier but also people working in the back to cook the different foods in the assembly line

What was my role in the information system?

I was a customer in the information system. When doing this project, I turn into a researcher observing the different employees and their roles as well as the customers that come and order.

Method page 1

Field Study: 2/10/25 at 1:30pm

I went to the Chipotle on University Hill after the lunch rush

Pros: Less people were there so I could get a clear view of what was going on. Going at a weird time that was 1:30 also let me see the start and ongoing efforts from the employees to prepare for dinner service. It also helped that since I went during a late lunch period, people were still there eating and talking. A big plus as well was I got my food super quick and it let me eat and observe for a larger time

Cons: There were less orders for me to watch the employees how it worked. As well as a lot of the employees were not at their normal stations of serving food but more of preparing and some even took their breaks to eat.

Method page 2

How did I Observe

I observed by being a regular customer and doing what normal customers do. I also took photos at all the steps I could be without disturbing others. I observed employees from a distance and saw how they interacted and stationed themselves in the restaurant. I also noted down key features about the restaurant and drew a map from which I could envision the restaurant.

What data did I collect

I collected photos and drew a map. I tried to get an interview but the employees decided not to do said interview so I respected their wishes.

Research Data page 1

Introduction

I choose to present my data from what is seen at the start of when you go to the Chipotle in Boulder. I felt that this was the best way for people to see the information at hand.

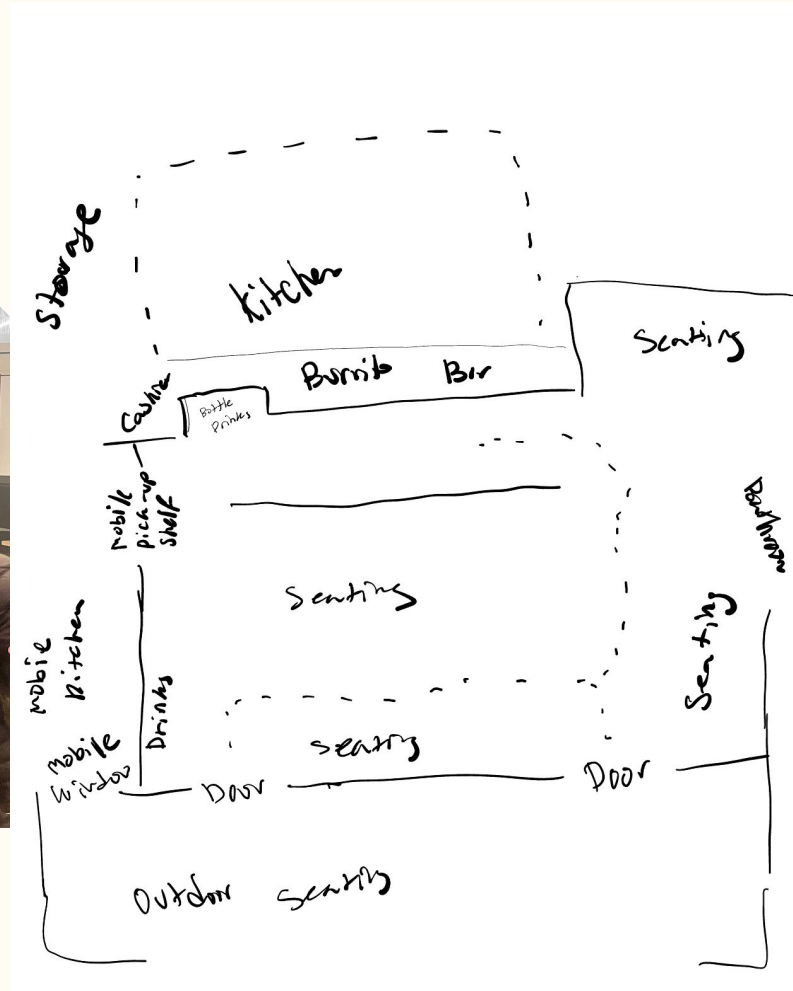


Research Data page 1.5

Map



Dragged my friend to Chipotle With me



Research Data page 2

Entrance

When you are about to enter the Chipotle, you can see that there are many features before you even walk in. There is an outdoor seating area that people can seat in. On the roof of the overhang, Chipotle has included heaters to let people eat outside when its colder in Boulder.

Seating Area

When walking in, you are greeted with a wide selection of inside seating. From seats against the window in the front to let people look outside to regular tables, it gives a wide variety of seating options for customers. The theme of the seating is a simple wood, metal area that is easily cleanable and looks uniform. Additionally, the seating area is strategically placed to funnel people into a single file line to order at the counter. I would suggest however to expand the area for the line. From prior experience when visiting, the Chipotle line will get quite large and the line space will go out the door. This can leave a bad experience for the customer and cause loss of business.



Research Data page 3

Mobile Pick Up Window

Positioned outside with the overhang, there is a mobile pick up window that employees can open and close to give customers their food. Customers will come and knock on the window and ask if their mobile orders are finished.



Mobile Order Station

With the mobile pick up window, there is also a smaller version of the larger burrito bar. This burrito bar is used to create said mobile orders which lets the restaurant not get backed up by mobile orders. This bar includes a tv/monitor that displays the different types of orders the employees will be making as well as a sticker label maker that Chipotle employees use to label the mobile orders and what is in it.



Research Data page 4

Burrito Bar

After lining up, individuals will see a large selection of options in different foods they are able to put into their burrito, quesadilla, burrito bowl, or tacos. Arranged in a straight line, the ingredients are ordered in the specific way in which you assemble said burrito. In this line, there are also stations that enable each employee to be in charge of a certain type of ingredient. There is a person doing the start of the burrito which include the rice, beans and the tortilla itself. Another person doing the meats and salsas, and then another person doing all the veggies and cheese. With this system, it streamlines the process of making the burrito as well as pushes people through the line faster by having multiple burritos being made. At the end, the burrito is wrapped in tin foil and workers use a sharpie with codes on it to let the person working the cashier know how much to charge the customer.

Overhanging Menu

On top of the burrito bar, there is an overhanging menu that list all the prices and ingredients you can get in your burrito. They also offer drinks, kids meals and sides such as chips as well on this menu.



Research Data page 5

Drink and Cashier

Right before the cashier, there is an assortment of bottled drinks here for the customer to grab. This is a great location as it lets people seamlessly grab a drink that they want before paying. At the Cashier station, there includes a card reader and below the cashier are the drink cups.

Fountain Drinks and Utensil Section

To the left of the cashier, there is a station that holds the fountain drink machine as well as the utensil stations. This station includes different sauces, forks, spoons, knives and napkins. Included in this section, there is a place to put trash and recycling.

Mobile Order Pick-Up

Directly between the Cashier and the Utensil Section, there is a few shelves that are designated for mobile orders. These include Doordash, Uber Eats and orders from their very own mobile ordering system.



Analysis page 1

Social Space: Actors

The actors in this information system are the customers, cashiers, cooks, and delivery pick up personal. There also is a manager that helps facilitate the employees of the restaurant.

Social Space: Division of Labor

1. The customer is directed in a line and is invited to come up to the cooks in the burrito bar to explain and tell them what they would like in their food before approaching the cashier to pay. They are prompted by the menu and the food seen in front of them when they are in line.
2. The cashier makes sure that the customer pays the correct amount for the food using the kiosk
3. Food chefs in the back make sure there is ample supply of food to be put into the food baskets in front of the cooks.
4. Cooks in the burrito bar assemble the food. They indicate if they are running low on something and interact with the customers. They also work at their own specific station to streamline the process
5. Mobile order cooks make the food for the mobile orders coming in.

Physical Space: Arrangement of People and Space

Refer to map above: Physical space is divided into division of labor which include main bar and cooks, mobile order bar and as well as a seating and line area for the customers. Employees are arranged in a order to streamline the process of making the burrito, passing said burrito from station to station in order to receive the food. The process of food making is very linear. Seating area is also arranged to streamline customers in a line to get said food. Customers interact with employees from start to finish of their food being made.

Physical Space: Arrangement of Artifacts

The space was not retrofit but very specific to what Chipotle wanted including customer flow, employee position as well as how food is assembled and created. Research Data was presented in order of dominant information flow where a new customer will walk in a see what is being presented.

Temporal Space

The space and the sequence of events that customers need to do in order to get their food seems almost like a manufacturing line. From walking in and lining up in a single file line to going through the burrito bar in a specific way, time is a very key factor in these decisions. The way the system works is that the creation of your burrito is supposed to be as fast as they can do it. To get you through the line and get you to eating. I was lucky to be there when someone brought a big order. They went item by item and got the order done super fast. This fast system lets them gain more profit and also gives customers satisfaction in getting their food as fast as possible.

Analysis page 4

Coordination

There is little discussion between cooks and cashiers besides using a sharpie to write down the code of what ingredients are in the burrito. Cooks use the sharpie to write down what is in the burrito like S for steak, G for guacamole, C for chicken and so on. Additionally, the cooks will sometimes use color plastic labels to help the cashier know if someone asked for double meat, double guac and so much more. All this communication is nonverbal.

There is however lots of communication between the customer and the employee to create the burrito. The verbal communication with asking for certain types of meats, extra meats, as well as different types of veggies and other things makes this interaction the most important.

There is also lots of communication between the cooks in the back and the cooks at the bar. Indicating what needs to be refilled and what is needed to start to cook for the next round of orders.

Propagation of Representational State of Information

In person order: Customer POV

Line up → verbal communication with first
cook about what type of food they
want.

- Continue down the line telling what
foods wanted
- Cashier sees code and inputs
food choices into kiosk for pricing
- pay
- Receipt.

Deviation of Design Recommendations or Principles

1. Recommendations
 - a. My first recommendation would be to make the area for the line better. The seating makes people funnel a certain way yes but they are also weaving around the people that are sitting down to eat. Additionally, the line sometimes goes out the door making the customers standing outside. I would recommend to take out the larger table in the back and instead set it aside as space to line up for customers. Most people do not sit down to eat at Chipotle and even then the seating area is never full.
 - b. My second recommendation would be to include pricing for the assortment of bottled drinks. In the photo, there is no information on the price that they charge for the drinks. If this information was provided, it would help give the necessary information for the customer as well as provide more information for them to make the correct decisions on what they drink. It would also prevent returns if people do not like the price.